CO2xN: Porewater Methane

Spetmebr 2025 Samples

2025-10-29

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##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden D. Blakley
```

```
#set the run date & user name
sample_year <- "2025"
sample_month <- "September"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_files <- c("Raw Data/20250922_2M_SEPT_GHG.csv","Raw Data/20250925_2M_SEPT_GHG.csv","Raw Data/2025092


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/LTREB_2M_ShimadzuGC_QAQC_2025.csv"


# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xN_Porewater_CH4_202509.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrized
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrized (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- ""

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

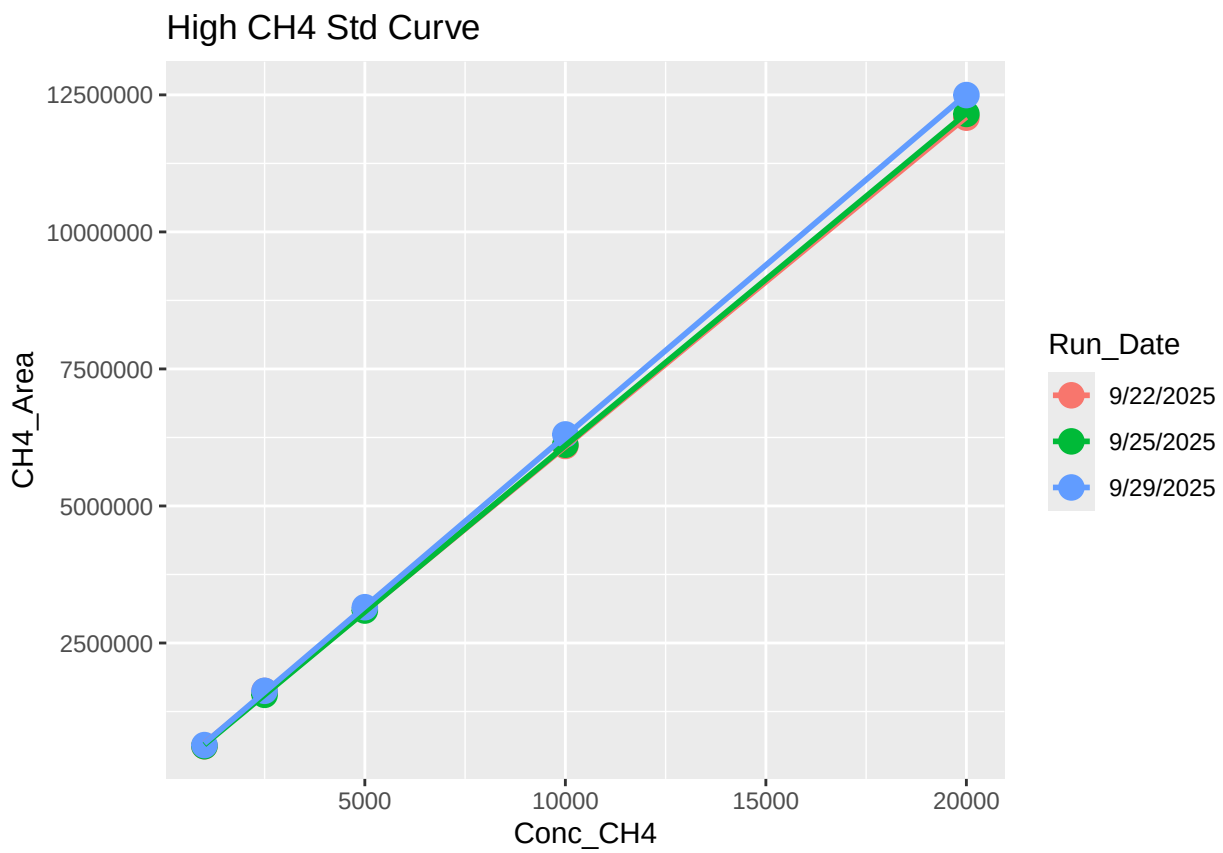
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

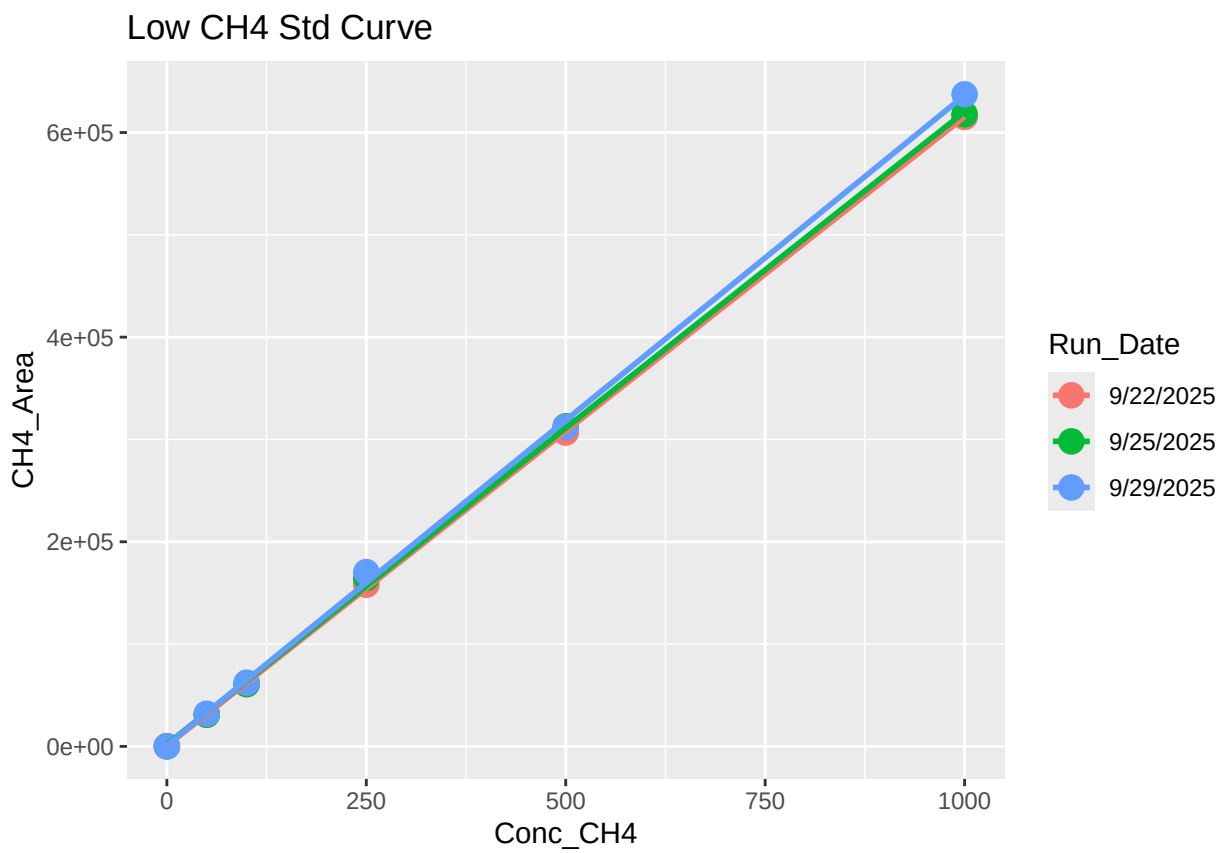
1 Read in data files

2 Assess standard curves

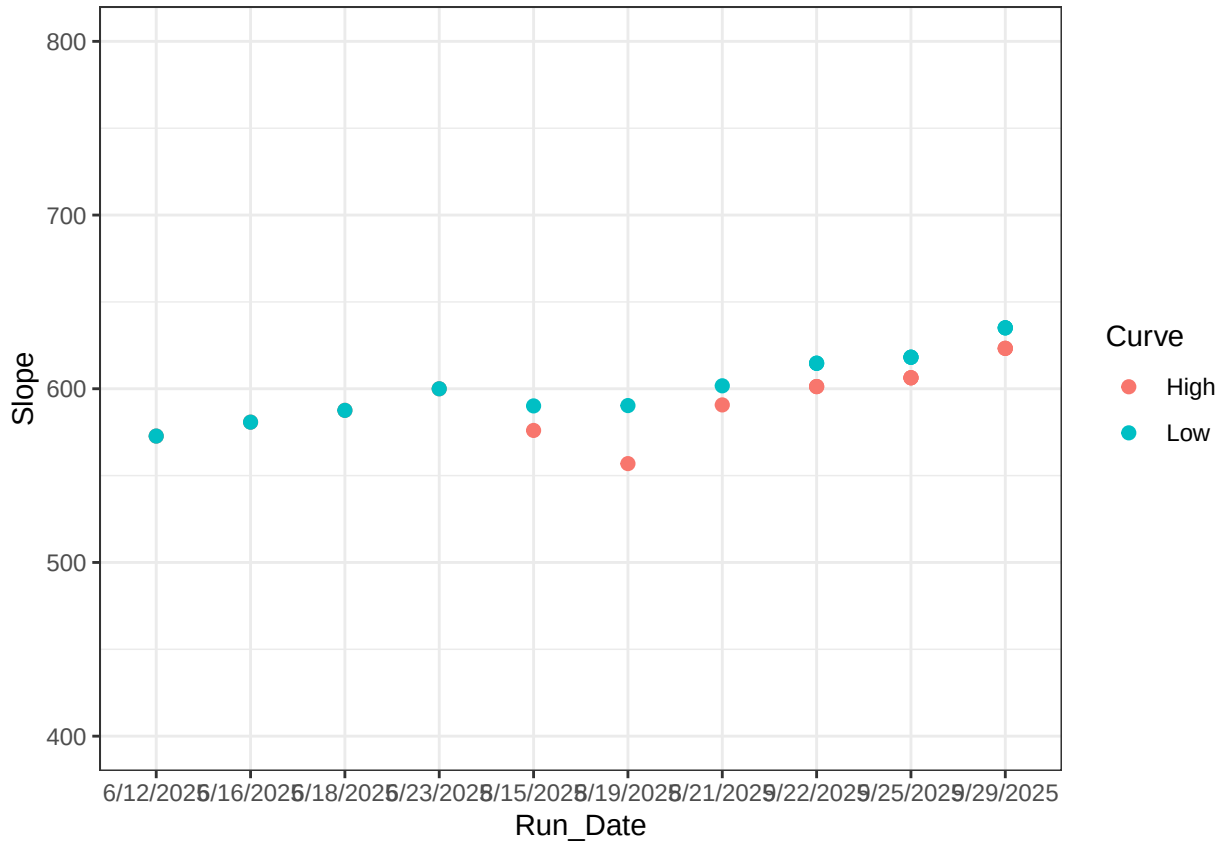
```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##   X  Run_Date Curve        R2    Slope Intercept
## 1 1 6/12/2025  High 0.9998670 572.7541 103940.35
## 2 2 6/16/2025  High 0.9988947 580.7674 116969.53
## 3 3 6/18/2025  High 0.9991820 587.5554 -62379.89
## 4 4 6/23/2025  High 0.9999761 599.9888  21668.58
## 5 5 6/12/2025   Low 0.9998670 572.7541 103940.35
## 6 6 6/16/2025   Low 0.9988947 580.7674 116969.53
```



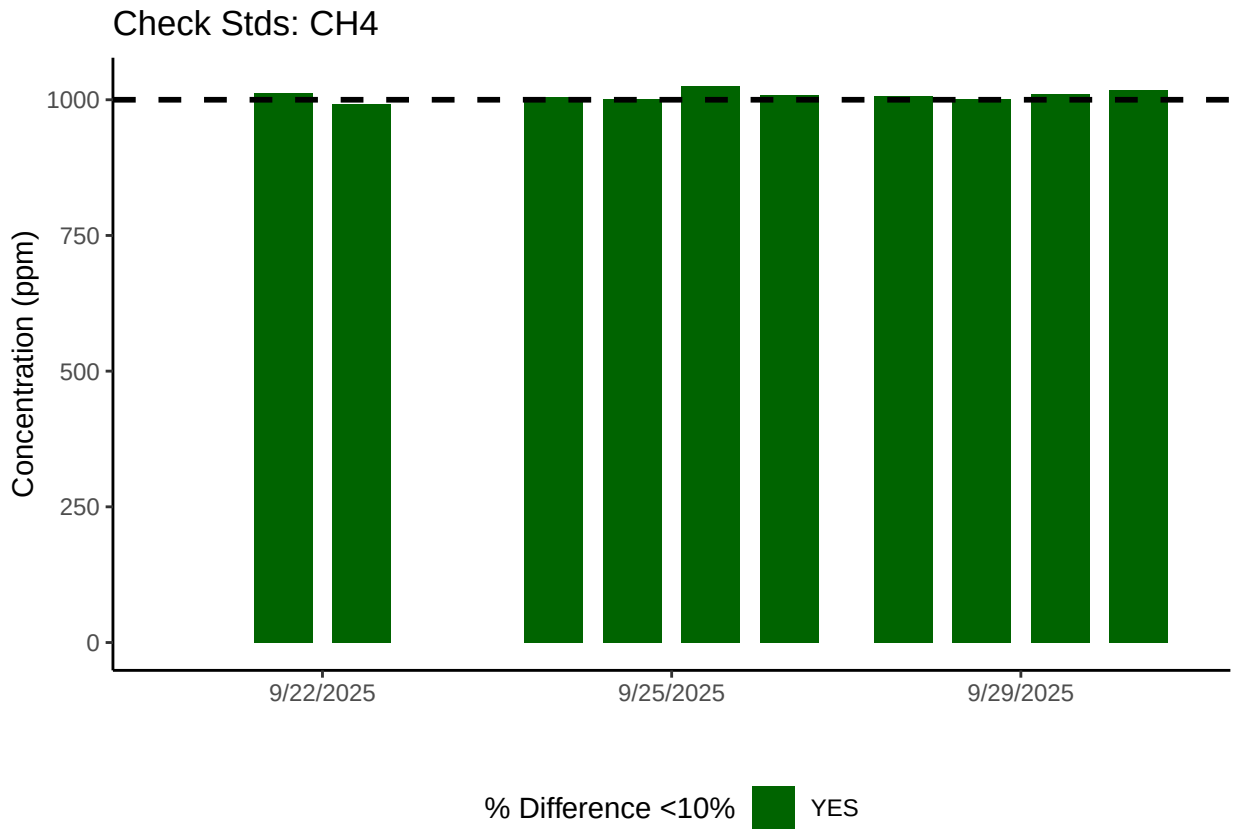

```
## [1] "High_R2 is Good-9/22/2025" "High_R2 is Good-9/25/2025"
## [3] "High_R2 is Good-9/29/2025"
```

```
## [1] "Low_R2 is Good-9/22/2025" "Low_R2 is Good-9/25/2025"
## [3] "Low_R2 is Good-9/29/2025"
```

2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

```
## Assess Blanks
```

```
## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED
```

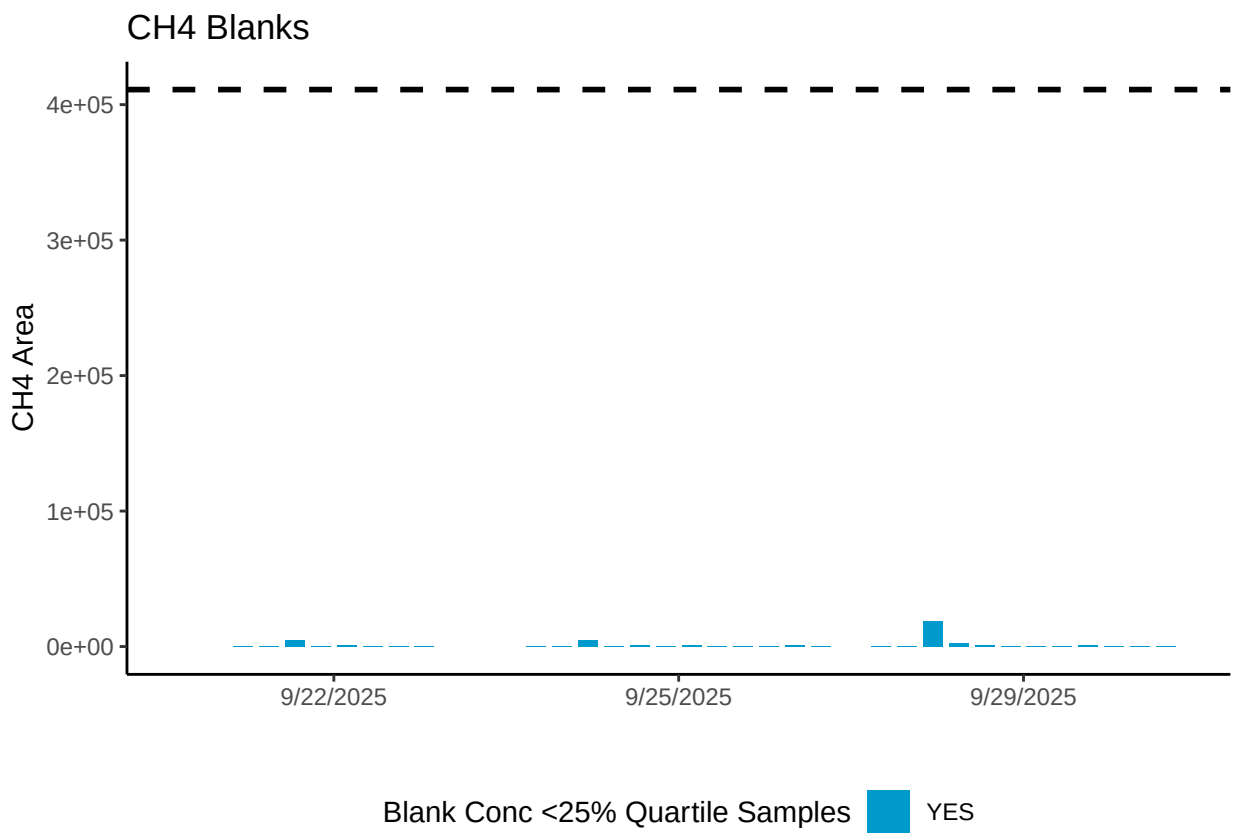
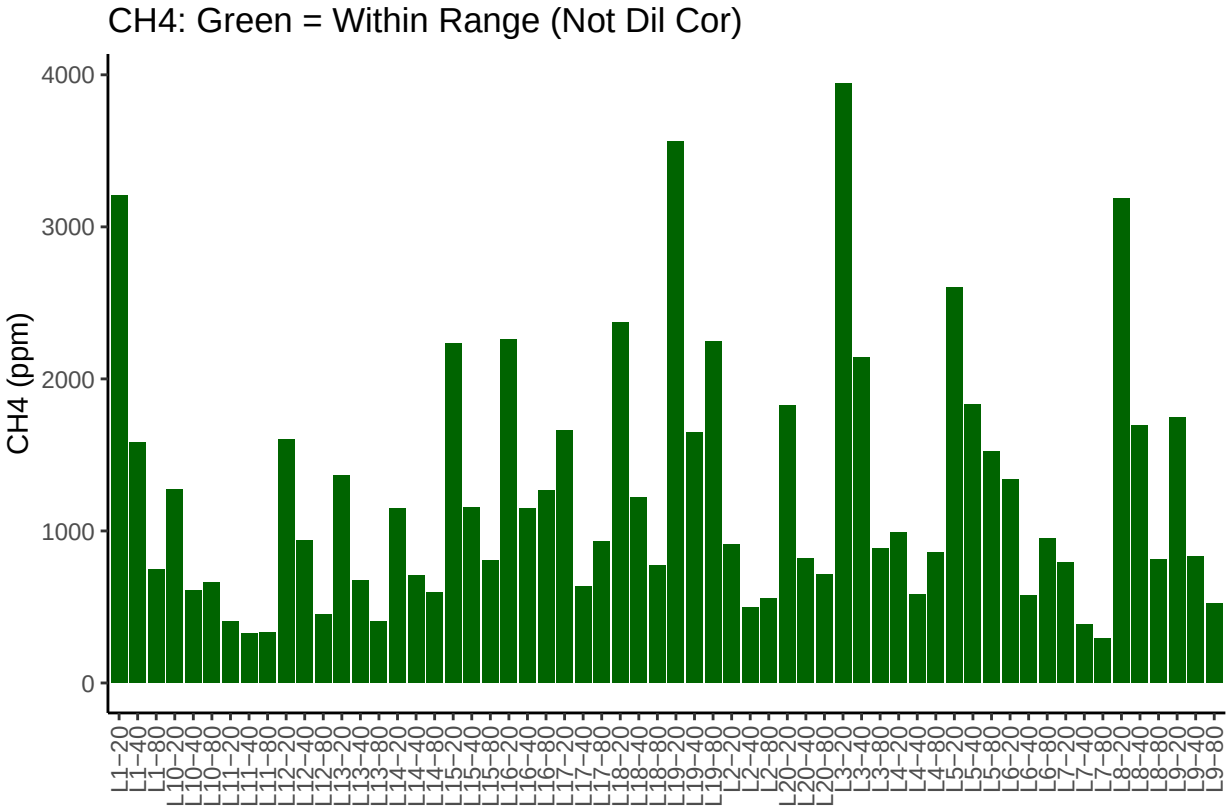
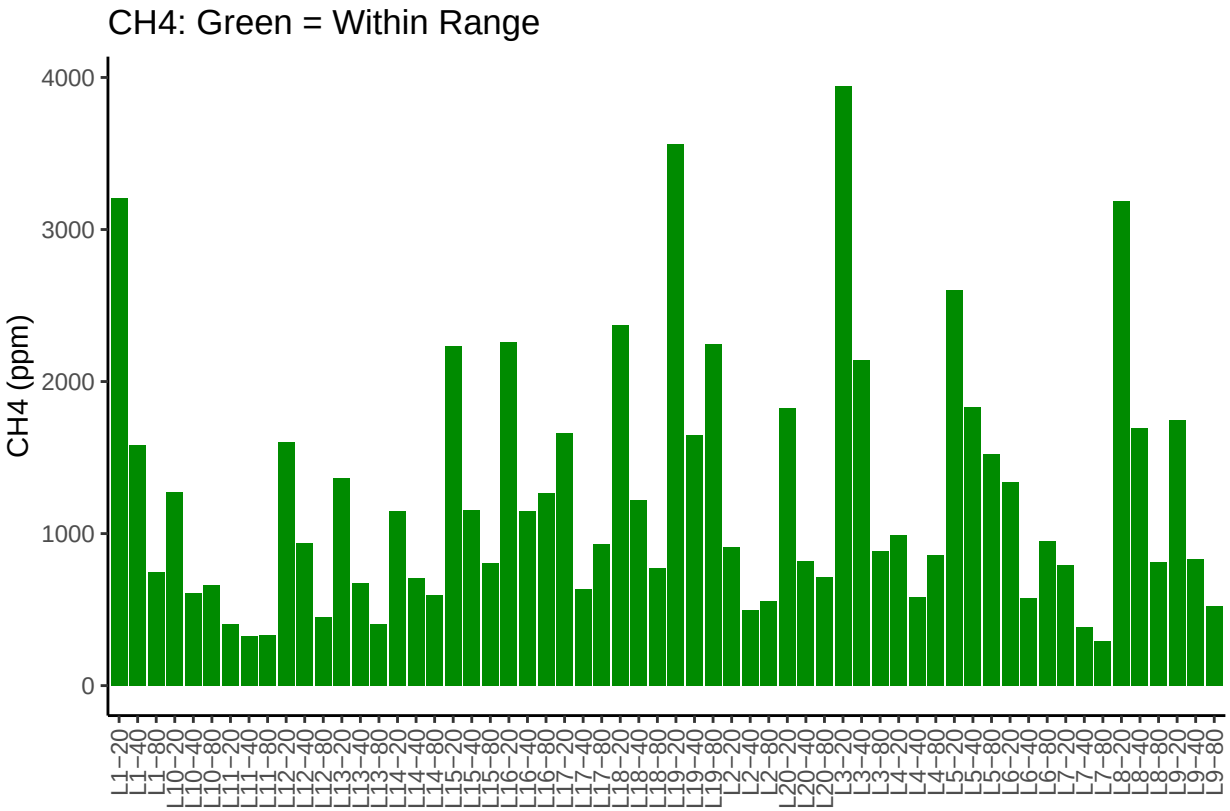


Table 1: Mean Area of Blanks

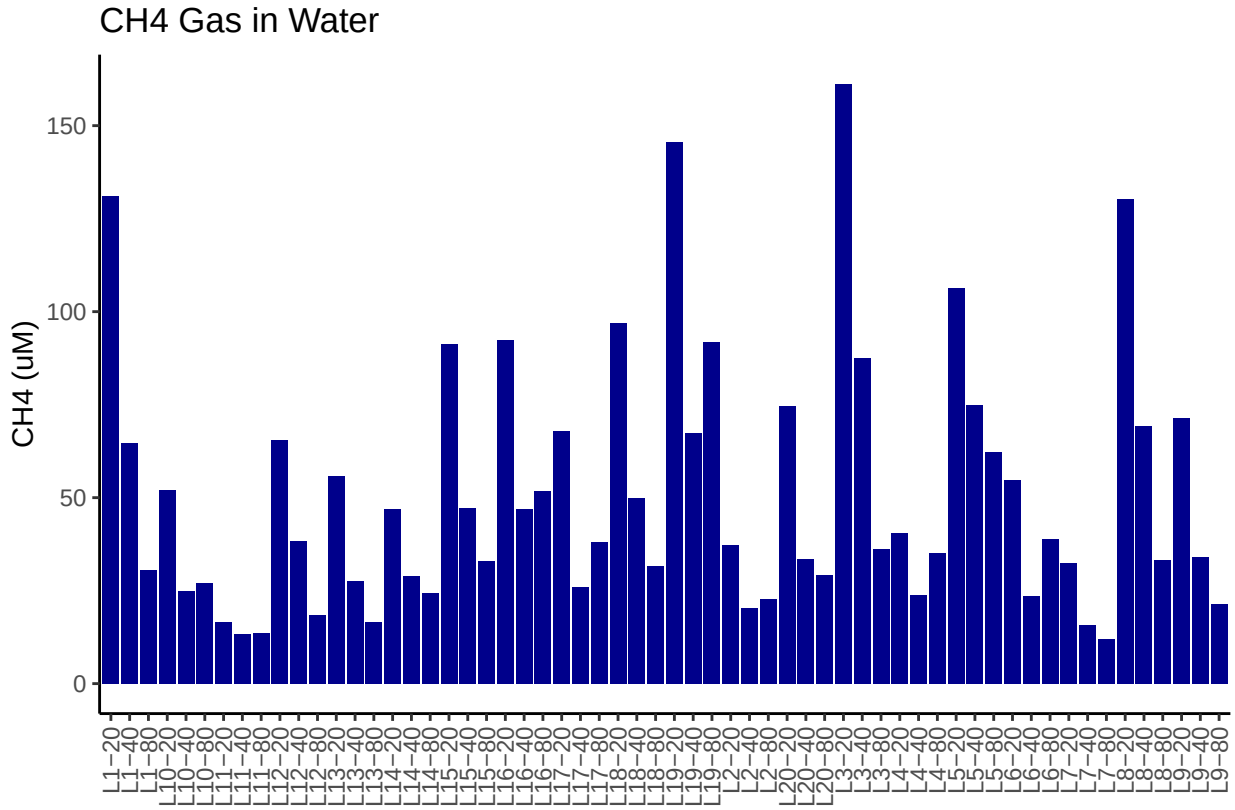
Test	Blank_Mean_Area
CH4	1334.531

5 Dilution correct samples and Check Range





6 Calculate gas in water



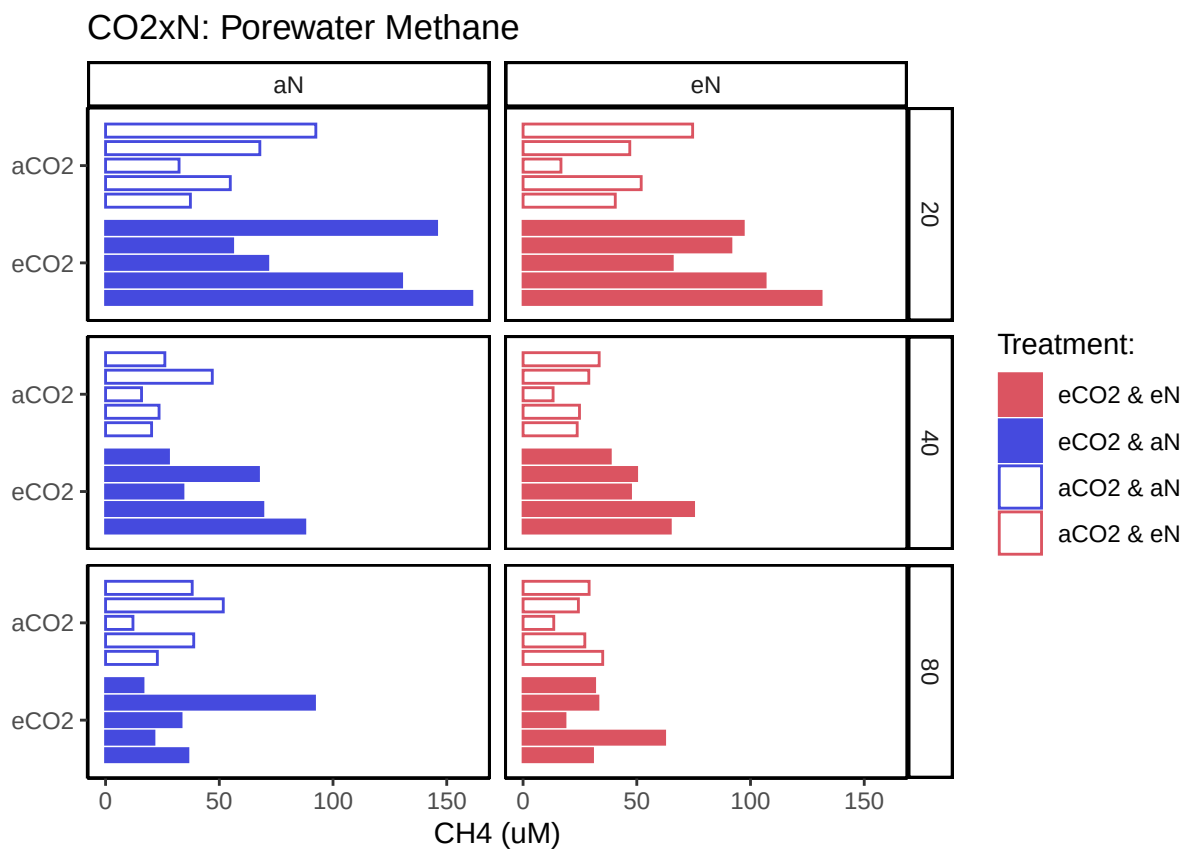
7 Merge samples with Metadata & check all samples are present

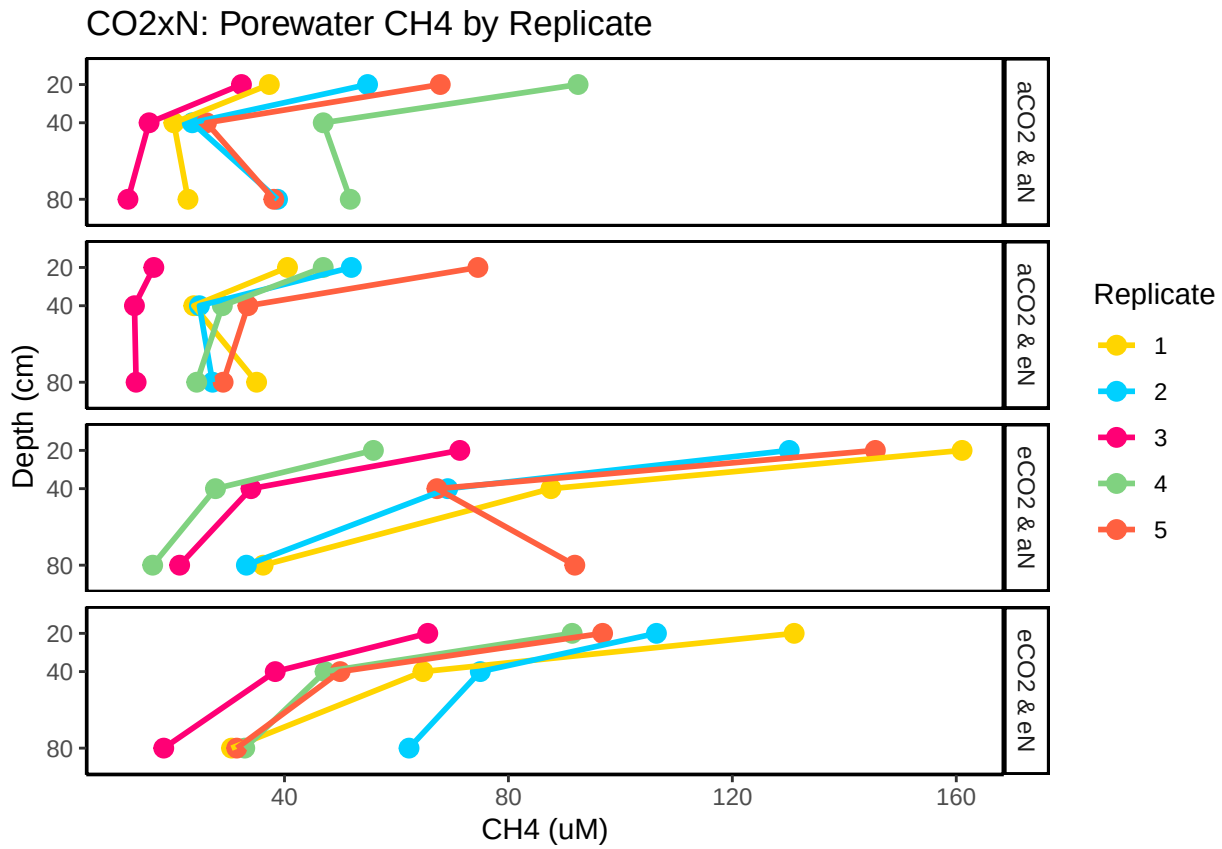
```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data
```

7.1 Organize Data

7.2 Visualize Data





##Export Data

#end